

Assignment 5

Due date: Apr 24th, 2026 before midnight

Problem 1. Consider the following discrete time Markov process $\{X_n; n \geq 0\}$ with the unit interval $[0, 1]$ as state space. Its transformation law is given by

$$X_{n+1} = \begin{cases} \frac{1}{3} + \frac{2}{3}X_n, & \text{with prob. } X_n \\ \frac{2}{3}X_n, & \text{with prob. } 1 - X_n \end{cases}$$

Show that this process is a martingale.

Problem 2. Let $\{X_n, n = 1, 2, \dots\}$ be a martingale. Prove that for any set of positive integers $k \leq l < m$ that the difference $X_m - X_l$ is uncorrelated with X_k , that is,

$$\mathbb{E}[(X_m - X_l)X_k] = 0.$$

Problem 3. Assume that $Y_1, Y_2, \dots$ are independent and identically distributed with $\mathbb{P}(Y_1 = 1) = p$, $\mathbb{P}(Y_1 = -1) = q = 1 - p$ and $p \neq q$. Fix positive integers a and b . With $S_0 = 0$, and $S_n = Y_1 + \dots + Y_n$, $n \geq 1$, let

$$T = \min\{n : S_n = -a \text{ or } S_n = b\}.$$

- (a) Verify that $M_n = S_n - n(p - q)$ and $M_n = (q/p)^{S_n}$ are martingales.
- (b) Find the probabilities $\mathbb{P}(S_T = -a)$ and $\mathbb{P}(S_T = b)$.
- (c) Find $\mathbb{E}[T]$.